

Knowing When to Abstain: Calibration and Selective Prediction for Machine-Learning Network Intrusion Detection

Anonymous (single author)
Affiliation withheld for review
author@example.org

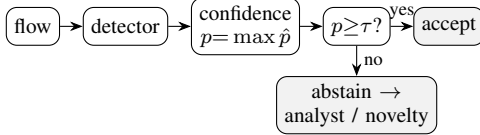


Fig. 1. The selective-prediction (reject-option) wrapper we evaluate: accept the detector’s label when its confidence p clears a threshold τ , otherwise abstain and defer to an analyst or a benign-only novelty stage. We ask whether p is trustworthy—and what becomes of attacks that look benign.

Abstract—A machine-learning intrusion detector that emits one hard label gives no signal of when to trust it and fails silently on attacks absent from training. We ask whether off-the-shelf detectors know when they are wrong—we propose no new detector. Under a controlled *leave-one-family-out* (LOFO) protocol on the public NSL-KDD benchmark, detection of a held-out attack family collapses (e.g. DoS from 0.863 to 0.603); a confidence-based reject option recovers only part of the gap, and almost none when the attack mimics benign traffic (R2L), which a benign-only novelty detector also cannot cleanly separate—a *structural* blind spot, not a tuning issue. The calibration measurements support this picture: the three standard detectors are markedly over-confident under the benchmark’s train–test shift (expected calibration error up to ≈ 0.22), post-hoc Platt scaling fitted on source data does not transfer, and split-conformal coverage falls from 0.930 ± 0.001 to 0.611 ± 0.002 . Selective prediction nonetheless cuts selective risk from ≈ 0.20 to ≈ 0.11 at 80% coverage—but only for models whose confidence ranks errors (tree ensembles, not logistic regression: area under the risk-coverage curve ≈ 0.05 versus ≈ 0.23). On a second, modern corpus (CIC-IDS-2017/2018) the within-distribution task is easy and well calibrated, yet an unknown family is still missed under cross-day and cross-corpus drift—so over-confidence is task-dependent while the abstention blind spot is not. Selective prediction is a cheap, model-agnostic safety layer for ML intrusion detection, with clearly characterised limits.

Index Terms—intrusion detection, selective prediction, calibration, uncertainty, open-set recognition, NSL-KDD

I. INTRODUCTION

A network intrusion detector that emits only “attack” or “benign” gives an operator no way to tell a confident verdict from a guess. In practice this matters twice over: alerts compete for scarce analyst attention, and genuinely novel attacks—absent from any training corpus—are exactly the cases a static classifier is least equipped to label. A detector that could say “I am not sure” on these inputs, and defer them to a human or

a secondary control (Fig. 1), would be more useful than one forced to commit.

This paper asks a narrow, practical question: do off-the-shelf machine-learning intrusion detectors know when they are wrong—producing confidence scores trustworthy enough to drive an abstention policy, and behaving safely on attacks they have never seen? *We propose no new detector; we evaluate existing ones.* We answer empirically on NSL-KDD [1], whose published test split deliberately contains attack types absent from training, and confirm the phenomena on a modern flow corpus (CIC-IDS-2017/2018) under cross-day and cross-corpus drift (Section IV-D). Our contributions are:

- We introduce a controlled *leave-one-family-out* (LOFO) open-set protocol—released as anonymized, reusable code (open-licensed at camera-ready)—and show that detection of a held-out attack family collapses, that a confidence reject option covers only part of the gap, and that attacks mimicking benign traffic (R2L) form a *structural* blind spot, missed by abstention and by a benign-only novelty stage alike (Section IV-C).
- We show selective prediction [2], [3] cuts selective risk substantially, but only for models whose confidence ranks their errors—tree ensembles, not the linear model—and that split-conformal coverage erodes under the same shift (Section IV-B).
- As the mechanism behind both, we measure that the three standard detectors are over-confident under distribution shift and that post-hoc Platt scaling [4] fitted on source data does *not* transfer (Section IV-A).

Throughout we follow the evaluation guidance of Arp *et al.* [5]: the calibrator is fit only on held-out training data, and no test information leaks into model selection. Unless stated otherwise, NSL-KDD test metrics are reported as the mean over $K=15$ seeds—resampling both the fit/calibration split and model randomness—with 95% confidence intervals (the canonical test partition is held fixed, so these intervals reflect train/model variance); equivalence is assessed with the *paired* per-seed difference and its CI rather than overlapping marginal intervals.

II. RELATED WORK

Calibration. Modern classifiers often output probabilities that do not match empirical accuracy [6], [7]. Expected cali-

bration error (ECE) summarises this gap, and temperature/Platt scaling are common post-hoc fixes [4]. These analyses typically assume the evaluation data is drawn from the training distribution—an assumption that fails for intrusion detection.

Selective prediction. A classifier with a reject option [2] trades coverage for accuracy on what it does accept; the risk–coverage curve and its area [3] quantify the trade-off. Maximum softmax/posterior probability is a strong baseline confidence score [8].

ML for intrusion detection. The limits of learned detectors—especially on novel attacks and under distribution shift—were argued early [9] and remain a methodological hazard [5]. We do not propose a new detector; we ask whether existing ones know when they are wrong.

Unknown attacks and shift. The failure of closed-world classifiers on inputs unlike their training data motivates out-of-distribution and open-set detection, for which the maximum softmax probability is the standard baseline [8]. NSL-KDD operationalises this within a security benchmark: its test split deliberately contains attack types absent from training [1], so measuring behaviour on them needs no synthetic construction. Our family-holdout protocol complements that built-in novelty with a controlled version that makes the “unknown” condition causal. We deliberately reuse only off-the-shelf scores and standard metrics so that the conclusions concern the *evaluation* of detectors, not a new detector design.

III. METHODOLOGY

Data. NSL-KDD [1] provides 125973 training and 22544 test records with 41 features (three categorical) and a binary normal/attack label; each attack maps to one of four families (DoS, Probe, R2L, U2R). Its test split contains 17 attack *types* (3750 records) that never appear in training—a built-in open-set test. Numeric features are standardised and categorical features one-hot encoded; the encoder is fit on training data only.

Models. We use scikit-learn [10] logistic regression, a random forest, and histogram gradient boosting—deliberately standard detectors, since our subject is their confidence, not their architecture.

Calibration. For each model we report ECE (15 equal-width bins of the predicted class probability) before and after Platt scaling, the latter fitted on a 20% held-out split of the training set.

Selective prediction. The confidence of a prediction is its maximum class probability. Sorting the test set by confidence and accepting the most-confident fraction c yields the selective risk $r(c)$; we report the area under $r(c)$ (AURC) and $r(0.8)$.

Open-set protocol (LOFO). To probe genuinely unknown attacks under control, our leave-one-family-out protocol removes one attack family entirely from training (keeping benign traffic and the other families), retrains, and measures detection and confidence on that family’s test records. We also report, at a global threshold giving 80% coverage, the fraction of unknown-family records that abstention rejects. Reported

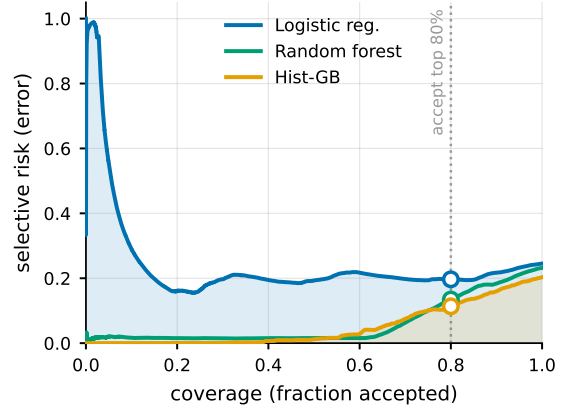


Fig. 2. Risk–coverage on NSL-KDD; the ring marks the 80%-coverage operating point and the shaded areas are the AURC (area under the curve) for the two extremes. Accepting only the most-confident inputs lowers selective risk sharply for the tree ensembles but barely for logistic regression, whose confidence does not rank its errors (nearly flat, and worst at the very top).

metrics are means over multiple seeds with 95% confidence intervals.

IV. RESULTS

A. Detectors are over-confident

All three detectors are markedly over-confident on the benchmark’s shifted test set, with ECE from ≈ 0.16 (random forest) to ≈ 0.22 (logistic regression) (Table I). Crucially this is induced by distribution shift, not intrinsic: on a held-out *in-distribution* split the same models are nearly perfectly calibrated ($\text{ECE} \lesssim 0.003$), and Platt scaling keeps them there. It is only on the shifted test set—which introduces novel attacks—that ECE inflates (Fig. 4), and there post-hoc Platt scaling fit on source data cannot remove it—ECE stays essentially unchanged for gradient boosting and is slightly worse for the random forest. Calibrating to the source distribution does not certify probabilities once the deployed distribution moves (Fig. 3); this is the core reason we turn to abstention, which needs only a usable confidence *ranking*, not calibrated probabilities.

B. Abstention helps—if confidence is informative

Abstaining on the least-confident inputs lowers selective risk—but only when the model’s confidence ranks its errors (Fig. 2). For gradient boosting, rejecting the least-confident 20% of inputs lowers selective risk to 0.1130 ± 0.0047 (AURC 0.0530 ± 0.0027); the random forest is similar (AURC 0.0604 ± 0.0013). Logistic regression is the cautionary case: its confidence barely ranks its errors (AURC 0.2259 ± 0.0115), so abstention buys little (0.1955 ± 0.0008). The lesson is that selective prediction is only as good as the underlying confidence signal, which here depends strongly on the model family.

C. Leave-one-family-out (LOFO) open-set evaluation

Detection of an attack family collapses once that family is unknown at training time, and a confidence reject option only

TABLE I
NSL-KDD TEST METRICS: MEAN $\pm$ 95% CI OVER $K=15$ SEEDS (RESAMPLING THE FIT/CALIBRATION SPLIT AND MODEL RANDOMNESS). LOWER ECE AND AURC ARE BETTER.

Model	Acc.	ECE	AURC	$r(0.8)$
Logistic reg.	0.752 ± 0.001	0.2151 ± 0.0008	0.2259 ± 0.0115	0.1955 ± 0.0008
Random forest	0.769 ± 0.004	0.1630 ± 0.0036	0.0604 ± 0.0013	0.1328 ± 0.0026
Hist. grad. boost.	0.799 ± 0.004	0.1806 ± 0.0037	0.0530 ± 0.0027	0.1130 ± 0.0047

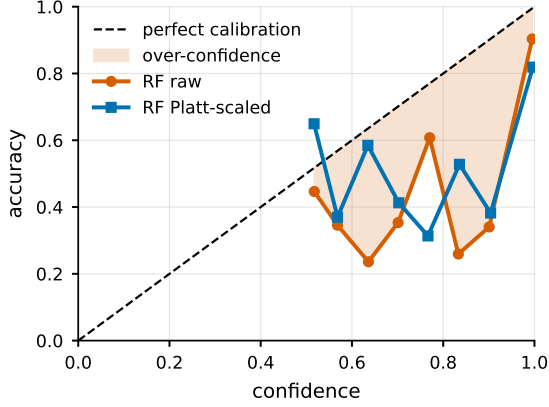


Fig. 3. Random-forest reliability on the shifted test set; the shaded wedge below the diagonal is the over-confidence (accuracy < confidence) that Platt scaling fitted on source data does not remove.

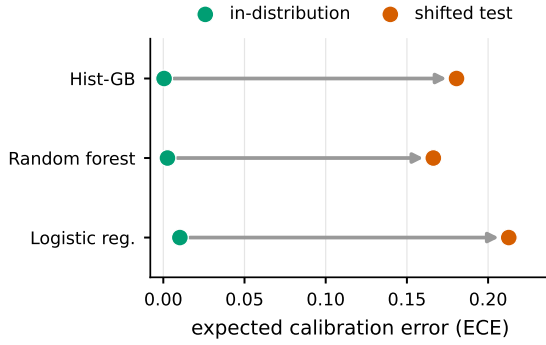


Fig. 4. Calibration inflates under shift: each arrow runs from a model’s in-distribution ECE (well calibrated) to its ECE on the shifted test set, where the novel attacks leave all three badly miscalibrated.

partly compensates. We establish this causally with a *leave-one-family-out* (LOFO) protocol: for each attack family F , remove all of F from training (keeping benign traffic and the other families), retrain, and measure detection and abstention on F ’s test records—released as anonymized, reusable code for review (open-licensed at camera-ready). On NSL-KDD’s built-in novel attacks the random forest flags ≈ 0.77 of *known* attacks but only ≈ 0.23 of novel ones; LOFO (Table II, Fig. 5) confirms this causally: detection of DoS falls from 0.863 to 0.603 when DoS is unknown, and Probe from 0.788 to 0.426, with abstention rejecting 0.406 and 0.371 at 80% coverage. The rare U2R family (train $n=52$, test $n=67$ —a high-variance estimate) behaves like the anomalous families: abstention still flags 0.61 of unknown U2R. R2L is the sharp exception—

TABLE II
LEAVE-ONE-FAMILY-OUT (LOFO) OPEN SET ON NSL-KDD: DETECTION WHEN A FAMILY IS SEEN IN TRAINING VS. HELD OUT (UNKNOWN), AND THE FRACTION OF UNKNOWN-FAMILY RECORDS ABSTENTION REJECTS AT 80% COVERAGE. MEANS OVER $K=10$ SEEDS; 95% CI HALF-WIDTH ≤ 0.064 (LARGEST FOR THE RARE U2R, $n=52/67$).

Family	det. (seen)	det. (unknown)	rejected by abstention
DoS	0.863	0.603	0.406
Probe	0.788	0.426	0.371
R2L	0.073	0.003	0.168
U2R	0.175	0.049	0.61

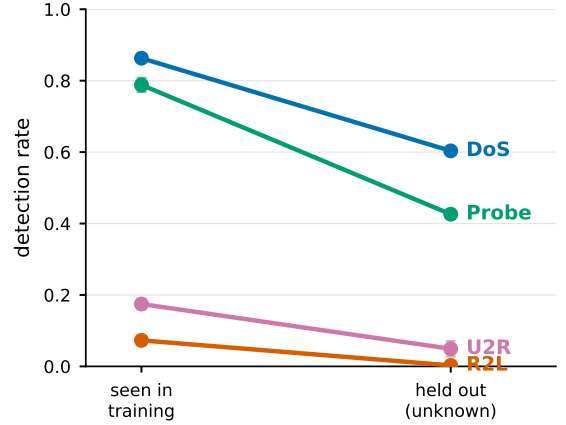


Fig. 5. Detection collapse under the LOFO protocol: each family’s detection rate slopes from seen-in-training to held-out (unknown). DoS/Probe/U2R fall steeply; R2L stays near zero even when seen—it mimics benign traffic, so there is little detection to lose.

detected at only 0.073 even when *seen*, 0.003 when unknown, with abstention flagging just 0.168. Because R2L traffic statistically resembles benign sessions the detector is confidently wrong, so the blind spot is specifically *benign-mimicking* attacks, not unknown attacks in general—R2L sits inside the benign cloud for two independent benign-only detectors alike (Fig. 6). The CIC-IDS-2017 cross-day experiment below is the same protocol on a second corpus, with the Web family held out.

D. Calibration is task-dependent; the blind spot is not (CIC-IDS-2017)

On an easier modern corpus the over-confidence disappears, yet an unknown family is still missed under drift—calibration is a property of the task, the abstention blind spot is not. We repeat the study on CIC-IDS-2017, a modern flow-based corpus (78 features), using three days (DDoS, PortScan, and

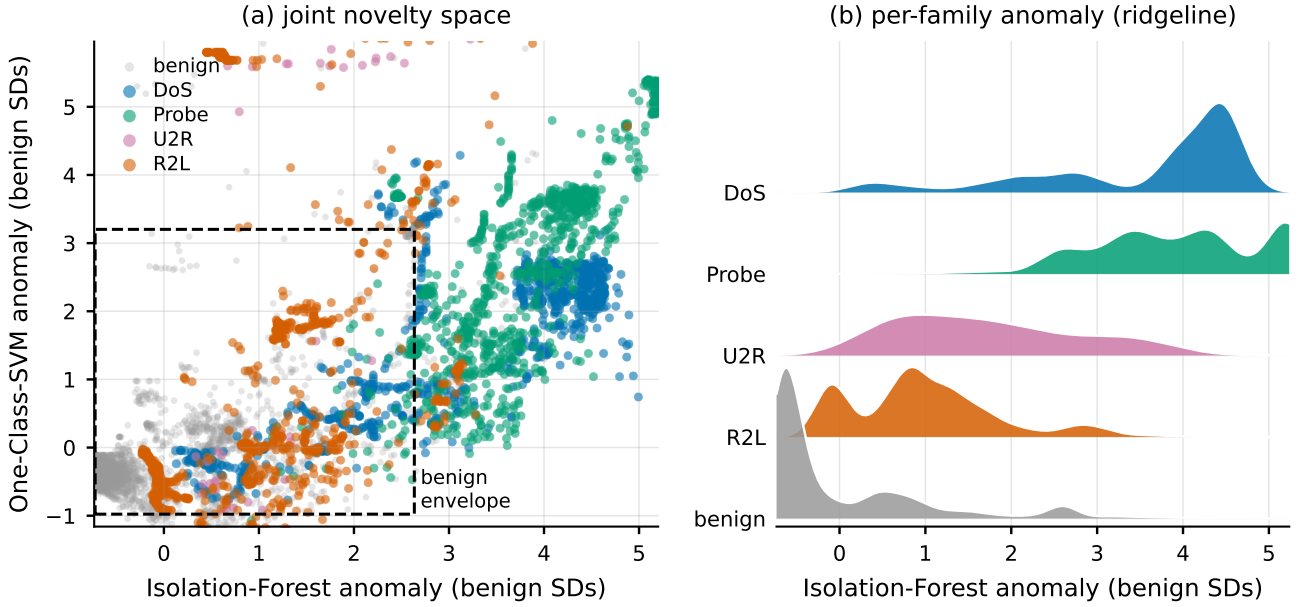


Fig. 6. Why R2L is the blind spot. (a) The joint novelty space: each NSL-KDD test record placed by how anomalous the two benign-only detectors find it, z-scored so benign sits near the origin (dashed box = central 95% of benign). (b) A ridgeline of the per-family Isolation-Forest anomaly. Both panels show R2L’s distribution sitting on top of benign’s—overlapping support—while DoS, Probe, and U2R separate, so neither abstention nor a benign-only novelty detector can cleanly flag the benign-mimicking R2L.

Web attacks). The picture is instructively different (Table III). Within a random split the task is far easier than NSL-KDD—gradient boosting reaches accuracy 0.9998 at ECE 0.0001—so over-confidence is *not* a problem here: calibration quality is a property of how hard and how shifted the task is, not an intrinsic flaw of the models. The value of abstention, however, persists under drift. We evaluate it with a cross-day split—training and testing on different days rather than a random mix—which avoids the optimistic temporal bias a random split would introduce [11]. Training on Friday (DDoS, PortScan) and testing on Thursday—whose Web-attack family is unknown and whose benign traffic has drifted—detection of Web falls from 0.9816 (seen) to 0.7916 (unknown); a confidence threshold at 80% coverage then rejects 0.9516 of those unknown-Web records, far more than the analogous NSL-KDD case because Web flows, unlike R2L, do not mimic benign traffic. The contrast across the two corpora is the substantive point: the calibration failure is task-dependent, while selective prediction is broadly useful—most when the unknown attack is statistically distinguishable from benign traffic.

Cross-corpus transfer. The strongest external-validity test trains on CIC-2017 and tests on CSE-CIC-IDS-2018 (benign + Infiltration), a different network and year, using the 27 CICFlowMeter features whose names match exactly across the two releases. The mechanisms reappear but degrade honestly under the real domain shift: accuracy falls from 0.96 (within-2017, same features) to 0.674, calibration collapses (ECE 0.2625), and the confidence signal weakens (AURC 0.2586 vs. 0.0043), so abstention recovers far less than within-corpus (selective risk 0.326→0.269 at 80% coverage). Infiltration, an

TABLE III
CIC-IDS-2017: REPLICATION (RANDOM SPLIT) AND CROSS-DAY DRIFT (TRAIN FRIDAY, TEST THURSDAY). MEANS OVER $K=5$ SEEDS; 95% CI HALF-WIDTH ≤ 0.0219 .

Model	Acc.	ECE	AURC	$r(0.8)$
Logistic reg.	0.9831	0.0157	0.0016	0.0017
Random forest	0.9998	0.0007	0.0	0.0
Hist. grad. boost.	0.9998	0.0001	0.0	0.0
Cross-day: Web detection 0.9816→0.7916 (unknown); abstention rejects 0.9516 of unknown Web at 80% coverage.				

unknown and benign-mimicking attack, is the cross-corpus blind spot: detected at only 0.142 and rejected by abstention at 0.232. Cross-corpus generalization is thus partial—the qualitative phenomena transfer, the quantitative guarantees do not (means over $K=5$ seeds; 95% CI half-width ≤ 0.0184).

E. Which uncertainty signal, and can we guarantee coverage?

Two questions sharpen the story: is maximum class probability (MSP) the right abstention signal, and can abstention carry a formal guarantee? For the first, we benchmark MSP against random-forest ensemble disagreement, Mahalanobis distance to class-conditional Gaussians, and k NN distance, on error ranking (AURC) and unknown-attack separation (Table IV). The appropriate statistic for a no-difference claim is the *paired* per-seed $\Delta\text{AURC} = \text{signal} - \text{MSP}$, and it shows that no signal dominates. Ensemble disagreement is statistically indistinguishable from MSP (paired $\Delta\text{AURC} - 0.0002 \pm 0.0003$; CI includes 0). Mahalanobis is marginally but significantly *better* at error ranking (-0.0022 ± 0.0014)

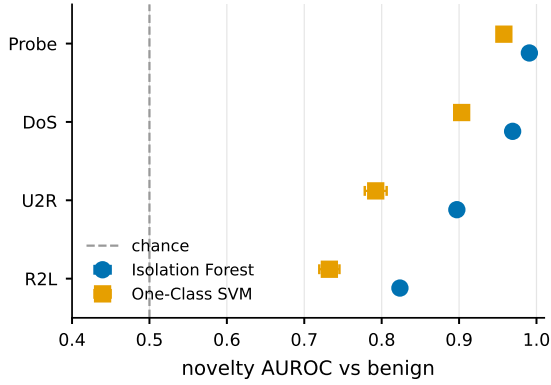


Fig. 7. Benign-only novelty detection (one-class models trained on benign records only): AUROC separating each attack family from benign, ordered by difficulty. DoS/Probe are easy; U2R/R2L—the abstention blind spot—are only partially recoverable, R2L the hardest. Markers are means over $K=8$ seeds, whiskers 95% CIs; the dashed line is chance.

yet significantly *worse* at unknown-attack detection (AUROC 0.807 ± 0.001 vs. MSP’s 0.881 ± 0.003); k NN is worse on both ($+0.0072 \pm 0.0030$). So contrary to the deep-learning out-of-distribution literature, where richer signals dominate under shift [12], [13], on tabular flow data the cheap max-softmax is a strong, well-rounded default—the leverage is in the abstention framework, not the estimator.

For the second, we wrap the detector in split conformal prediction [14], which gives distribution-free coverage under exchangeability. At target coverage 0.90, the prediction sets attain empirical coverage 0.930 ± 0.001 in-distribution—valid—but only 0.611 ± 0.002 under the train-test shift. This is the conformal analogue of the calibration failure of Section IV-A: the guarantee is exact under exchangeability and erodes precisely when the deployment distribution moves [15]. Restoring it needs shift-aware (weighted) conformal with estimable likelihood ratios—impractical for genuine zero-days, the honest caveat.

F. A benign-only novelty stage partially closes the blind spot

Confidence-based abstention barely flags R2L (Section IV-C) because such attacks lie on the benign feature manifold (Fig. 6). The apt detector for that regime is a *benign-only* density model—a one-class novelty detector, distinct from the two-class Mahalanobis used above for error ranking. We fit an Isolation Forest [16] and a one-class SVM [17] on benign records only. At a 10% benign false-positive budget the Isolation Forest recovers 0.513 of U2R and 0.243 of R2L (the one-class SVM 0.657 and 0.41; means over $K=8$ seeds, 95% CI half-width ≤ 0.036)—against abstention’s ≈ 0.168 for unknown R2L, a clear if partial gain. The AUROC ceilings (Fig. 7: R2L 0.824, U2R 0.897) show the limit is real but not absolute: R2L is hardest because its support most overlaps benign, so no benign-only detector fully recovers it, yet a novelty stage is a cheap, measured complement—abstention and novelty catch different failure modes (confident errors vs. benign-mimicking attacks).

V. DISCUSSION AND LIMITATIONS

Selective prediction is a cheap, model-agnostic safety layer: at the cost of deferring a minority of inputs it materially lowers the error among those acted upon, and it partially flags unknown attacks. But our results bound the claim. First, the value of abstention is inherited from the confidence signal, which is good for tree ensembles and poor for the linear model. Second, post-hoc calibration on source data does not survive distribution shift, so calibrated probabilities should not be trusted as deployment-time guarantees. Third, and most important operationally, abstention only catches unknowns that *look* anomalous; attacks that mimic benign traffic (R2L here) are missed confidently and silently.

Operational recipe. Deploy a model whose confidence is informative (low validation AURC—a tree ensemble here, not the linear model); set the operating coverage from the risk-coverage curve to meet a target precision; route the abstained minority to a secondary control; monitor the abstention rate as a cheap, label-free drift signal [18], [19]; and pair abstention with the benign-only novelty stage above for the benign-mimicking attacks it cannot catch.

Limitations. NSL-KDD is dated and its features hand-crafted, so absolute numbers will differ on modern corpora; the mechanisms we isolate—over-confidence under shift, model-dependent confidence quality, and the benign-mimicry blind spot—are not specific to it. We study binary detection with a single abstention signal (maximum class probability); richer signals and the multi-class attack-typing setting are natural extensions. We evaluate a cross-day split and a cross-corpus transfer (CIC-2017→2018) on shared features; a full time-ordered evaluation, the label-corrected CIC-2017 [20], and standardized NetFlow-v2 multi-network transfer (a common schema avoiding our 27-feature restriction) remain future work.

VI. CONCLUSION

On a hard benchmark (NSL-KDD) off-the-shelf intrusion detectors are over-confident and post-hoc calibration does not transfer under shift; on an easier modern corpus (CIC-IDS-2017) they are well-calibrated, so the need for calibration is task-dependent. Across both, a simple reject option lowers selective risk and flags unknown attacks—strongly when those attacks are distinguishable from benign traffic, weakly when they mimic it. Selective prediction is worth adding to ML intrusion detection, provided its confidence signal and its blind spots are measured rather than assumed.

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TABLE IV

UNCERTAINTY SIGNALS ON NSL-KDD (MEAN $\pm$ 95% CI OVER $K=15$ SEEDS). AURC LOWER IS BETTER, UNKNOWN-AUROC HIGHER; Δ AURC IS THE PAIRED PER-SEED DIFFERENCE VS. MSP (NEGATIVE = BETTER; A CI SPANNING 0 MEANS NO DETECTABLE DIFFERENCE).

Signal	AURC	unknown-AUROC	Δ AURC vs. MSP
Max-softmax (MSP)	0.0604 $\pm$ 0.0013	0.881 $\pm$ 0.003	—
RF ensemble disagreement	0.0602 $\pm$ 0.0013	0.881 $\pm$ 0.003	-0.0002 $\pm$ 0.0003
Mahalanobis distance	0.0581 $\pm$ 0.0012	0.807 $\pm$ 0.001	-0.0022 $\pm$ 0.0014
kNN distance	0.0676 $\pm$ 0.0032	0.830 $\pm$ 0.006	+0.0072 $\pm$ 0.0030

REFERENCES

- [1] M. Tavallaei, E. Bagheri, W. Lu, and A. A. Ghorbani, "A detailed analysis of the KDD CUP 99 data set," in *IEEE Symposium on Computational Intelligence for Security and Defense Applications (CISDA)*, 2009, pp. 1–6.
- [2] R. El-Yaniv and Y. Wiener, "On the foundations of noise-free selective classification," *Journal of Machine Learning Research*, vol. 11, pp. 1605–1641, 2010.
- [3] Y. Geifman and R. El-Yaniv, "Selective classification for deep neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4878–4887.
- [4] J. Platt, "Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods," in *Advances in Large Margin Classifiers*, A. J. Smola, P. Bartlett, B. Schölkopf, and D. Schuurmans, Eds. MIT Press, 1999, pp. 61–74.
- [5] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, "Dos and don'ts of machine learning in computer security," in *USENIX Security Symposium*, 2022, pp. 3971–3988.
- [6] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *International Conference on Machine Learning (ICML)*, 2017, pp. 1321–1330.
- [7] A. Niculescu-Mizil and R. Caruana, "Predicting good probabilities with supervised learning," in *International Conference on Machine Learning (ICML)*, 2005, pp. 625–632.
- [8] D. Hendrycks and K. Gimpel, "A baseline for detecting misclassified and out-of-distribution examples in neural networks," in *International Conference on Learning Representations (ICLR)*, 2017.
- [9] R. Sommer and V. Paxson, "Outside the closed world: On using machine learning for network intrusion detection," in *IEEE Symposium on Security and Privacy (S&P)*, 2010, pp. 305–316.
- [10] F. Pedregosa et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [11] F. Pendlebury, F. Pierazzi, R. Jordaney, J. Kinder, and L. Cavallaro, "TESSERACT: Eliminating experimental bias in malware classification across space and time," in *USENIX Security Symposium*, 2019, pp. 729–746.
- [12] Y. Ovadia, E. Fertig, J. Ren, Z. Nado, D. Sculley, S. Nowozin, J. V. Dillon, B. Lakshminarayanan, and J. Snoek, "Can you trust your model's uncertainty? Evaluating predictive uncertainty under dataset shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [13] K. Lee, K. Lee, H. Lee, and J. Shin, "A simple unified framework for detecting out-of-distribution samples and adversarial attacks," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2018, pp. 7167–7177.
- [14] A. N. Angelopoulos and S. Bates, "Conformal prediction: A gentle introduction," *Foundations and Trends in Machine Learning*, vol. 16, no. 4, pp. 494–591, 2023.
- [15] R. J. Tibshirani, R. Foygel Barber, E. J. Candès, and A. Ramdas, "Conformal prediction under covariate shift," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2019, pp. 2526–2536.
- [16] F. T. Liu, K. M. Ting, and Z.-H. Zhou, "Isolation forest," in *IEEE International Conference on Data Mining (ICDM)*, 2008, pp. 413–422.
- [17] B. Schölkopf, J. C. Platt, J. Shawe-Taylor, A. J. Smola, and R. C. Williamson, "Estimating the support of a high-dimensional distribution," *Neural Computation*, vol. 13, no. 7, pp. 1443–1471, 2001.
- [18] R. Jordaney, K. Sharad, S. K. Dash, Z. Wang, D. Papini, I. Nouretdinov, and L. Cavallaro, "Transcend: Detecting concept drift in malware classification models," in *USENIX Security Symposium*, 2017, pp. 625–642.
- [19] L. Yang, W. Guo, Q. Hao, A. Ciptadi, A. Ahmadzadeh, X. Xing, and G. Wang, "CADE: Detecting and explaining concept drift samples for security applications," in *USENIX Security Symposium*, 2021, pp. 2327–2344.
- [20] G. Engelen, V. Rimmer, and W. Joosen, "Troubleshooting an intrusion detection dataset: The CICIDS2017 case study," in *IEEE Security and Privacy Workshops (SPW)*, 2021, pp. 7–12.